Reviewer Pack — Minimal Rank of Arithmetic Progression Solutions vs ACC⁰ Low…

Entry 4d89feedec6f · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Arithmetic Progression Solutions vs ACC⁰ Lower Bounds

Entry ID: `4d89feedec6f` **Verdict**: `INCONCLUSIVE` **Recorded**: 2026-05-26 18:39:28 UTC

1. Conjecture

Field A (mathematical branch): Number Theory (Arithmetic Progressions) **Field B** (complexity object): Complexity Theory: ACC⁰ Lower Bounds for Explicit Functions

Statement:

> ['For every explicit function f in P, there exists a constant C such that the minimal rank of the set of arithmetic progressions that are solutions to the system of equations defined by f is at least C·log(n).', 'Equivalently, there exists an algorithm that, given an explicit function f in P, can determine whether this minimal rank is less than some threshold c·log(n) in subexponential time.', 'For all explicit functions f in P, if f has a degree greater than 2, then the minimal rank of the set of arithmetic progressions that are solutions to the system of equations defined by f is at least C·log(n).']

Rationale (proposer's reasoning):

> ['Arithmetic progressions provide a rich source of algebraic structures that can be used to encode complexity. Their study in number theory has potential applications to complexity theory.', 'This conjecture links arithmetic progressions, which are common in number theory, with ACC⁰ lower bounds, an area of complexity theory. If true, it would provide a new tool for proving lower bounds on explicit functions.', 'The minimal rank of arithmetic progressions as solutions to equations is computationally accessible and could be used to derive lower bounds on the complexity of explicit functions.']

Taxonomy category: `ACC_SIPSER` (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): `af869ca6cc347c63`

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

> If the mean minimal rank across all functions in P with degrees up to 2 is at least C·log(n) and no seed produces a minimal rank less than 0.5·C·log(n), then the conjecture is supported.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM_1	LLM_2
RELATIVIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: ADJACENT_OK against 9 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - 'arithmetic progression' AND 'minimal rank' AND 'ACC⁰ lower bounds' - 'number theory' AND 'complexity theory' AND 'degree greater than 2' AND 'subexponential time' - 'solutions to equations' AND 'explicit function' AND 'logarithmic lower bound'

Top relevant hits considered: - [http://arxiv.org/abs/0706.3841v1] Arithmetic lattices and weak spectral geometry - [http://arxiv.org/abs/2410.10707v3] Cusp types of arithmetic hyperbolic manifolds - [http://arxiv.org/abs/2410.15580v2] Language Models are Symbolic Learners in Arithmetic - [http://arxiv.org/abs/1311.1421v3] Multiplicative differential algebraic K-theory and applications - [http://arxiv.org/abs/2304.14284v3] Torsion primes for elliptic curves over degree 8 number fields - [http://arxiv.org/abs/cs/9811005v1] Writing and Editing Complexity Theory: Tales and Tools - [http://arxiv.org/abs/1911.11077v2] Asymptotic expansions with exponential, power, and logarithmic functions for non-autonomous nonlinear differential equat - [http://arxiv.org/abs/1810.10273v2] Accurate and efficient explicit approximations of the Colebrook flow friction equation based on the Wright-Omega functio

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)
Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution**: rc=1, elapsed=0.1s

5.1 Generated Python source

```python
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_polynomial(degree):
        coefficients = [random.randint(1, 10) for _ in range(degree + 1)]
        return coefficients

    def find_arithmetic_progressions(poly, n):
        progressions = []
        for a in range(-n, n+1):
            for d in range(-n, n+1):
                if d == 0:
                    continue
                progression = [a + i * d for i in range(n)]
                if all(poly[i] == sum(coeff * x**i for coeff, x in zip(poly, progression)) for i i
                    progressions.append(progression)
```

```
        return progressions

    def minimal_rank(progressions):
        rank = len(set(tuple(p) for p in progressions))
        return rank

    n_values = [5, 10, 15, 20, 30, 40]
    total_rank = 0
    instances_tested = 0

    for n in n_values:
        poly = generate_polynomial(random.randint(1, 2))
        progressions = find_arithmetic_progressions(poly, n)
        rank = minimal_rank(progressions)
        total_rank += rank
        instances_tested += len(progressions)

    mean_value = total_rank / instances_tested

    C = 1.0
    threshold = 0.5 * C * math.log(n_values[-1])

    conjecture_holds = mean_value >= threshold
    counterexample = "" if conjecture_holds else f"rank={mean_value}, expected={threshold}"

    return {
        "metric_name": "minimal_rank",
        "metric_value": mean_value,
        "instances_tested": instances_tested,
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }

if __name__ == "__main__":
    seeds = [int(x) for x in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3
    results = []

    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std=0.0 support_fraction=1.0")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_value} std=0.0 support_fraction={support_fraction}")
    else:
        first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"rank too low\" first_failing_seed={first_failin
```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d12e0b78.py", line 72, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d12e0b78.py", line 51, in run_trial
    mean_value = total_rank / instances_tested
                 ~~~~~~~~~~~^~~~~~~~~~~~~~~~~~
ZeroDivisionError: division by zero
```

8. Critic adversarial review

Critic verdict: `CHALLENGE`

Critic reasoning:

> skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: `INCONCLUSIVE`

Reasoning:

> The test crashed before producing data, which means it did not complete its execution to provide a result. | next: Re-run the test with appropriate error handling to ensure it completes and provides a result.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	12766
2	preregistration	ollama_remote	glm4:latest	0	0	5561
3	novelty	ollama_remote	glm4:latest	0	0	4899
4	novelty	ollama_remote	glm4:latest	0	0	6095
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	42953
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	5288
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8705
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8202
9	judge	ollama_remote	glm4:latest	0	0	8620

Totals: 0 input tokens, 0 output tokens, ~$0.0000 cost, 103088 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/4d89feedec6f.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice

3. The full LLM transcripts are in `research/audit/4d89feedec6f.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)
Replay tarball: `research/replay/4d89feedec6f.tar.gz` (if generated)